

U THONG AGROMET, Thailand

WMO: 484270

Lat: 14.304N

Lon: 99.865E

Elev: 7

StdP: 101.24

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.4	(c) 18.1	(d) 11.4	(e) 8.4	(f) 20.0	(g) 13.6	(h) 9.7	(i) 21.8	(j) 4.7	(k) 25.8	(l) 4.0	(m) 24.9	(n) 0.2	(o) N/A	(p) 0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 10.0	(c) 38.0	(d) 25.7	(e) 36.9	(f) 25.7	(g) 35.9	(h) 25.8	(i) 27.7	(j) 32.8	(k) 27.4	(l) 32.5	(m) 27.1	(n) 32.3	(o) 0.8	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.2	29.5	26.2	21.7	29.1	26.0	21.4	28.8	88.4	33.0	86.9	32.4	85.7	32.5	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.1	3.1	2.4	14.1	39.1	2.4	1.3	12.3	40.0	10.9	40.8	9.6	41.5	7.8	42.4		
			12.7	29.0	2.4	1.0	11.0	29.7	9.6	30.3	8.2	30.8	6.5	31.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	25.2	27.5	29.4	31.1	30.7	29.9	29.3	29.2	28.9	28.3	27.3	25.4
	DBStd	2.44	2.34	1.79	2.07	1.81	1.63	1.16	1.23	1.00	1.01	1.09	1.50	2.28
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6756	472	489	603	633	642	596	597	595	567	567	518	477
	CDD18.3	3714	213	256	344	383	384	346	339	337	317	309	268	219
	CDH23.3	42297	1910	2765	4268	5124	4939	4227	3953	3843	3496	3285	2618	1869
	CDH26.7	19175	718	1283	2211	2877	2566	1940	1692	1602	1409	1242	974	661
Wind	WSAvg	0.4	0.3	0.3	0.4	0.4	0.4	0.5	0.5	0.5	0.3	0.2	0.4	0.5
Precipitation	PrecAvg	1071	7	14	36	66	137	101	108	113	232	208	43	6
	PrecMax	1397	53	92	175	233	276	192	184	184	434	366	132	40
	PrecMin	743	0	0	0	1	17	12	50	52	104	29	0	0
	PrecStd	153	11	21	40	54	59	42	40	34	85	81	41	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	35.6	38.3	39.8	39.2	37.0	36.1	35.6	35.1	34.4	33.6	33.5
		MCWB	24.8	25.4	24.5	25.0	25.9	26.3	25.8	26.2	26.5	26.5	26.4	25.1
	2%	DB	32.8	34.6	36.8	38.4	37.5	35.6	34.7	34.3	34.1	33.2	32.9	32.8
		MCWB	24.8	25.1	24.7	25.8	26.1	26.3	25.8	26.0	26.5	26.4	26.0	24.9
	5%	DB	31.8	33.7	35.8	37.4	36.3	34.5	33.7	33.4	33.3	32.6	32.2	31.7
		MCWB	24.4	24.8	25.0	25.8	26.3	26.2	25.7	25.9	26.3	26.4	25.6	24.6
	10%	DB	30.6	32.8	34.8	36.1	35.1	33.4	32.7	32.5	32.4	31.8	31.4	30.4
		MCWB	23.8	24.6	25.2	25.8	26.3	26.1	25.6	25.8	26.2	26.2	25.1	23.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	27.1	27.5	27.9	28.2	28.1	27.2	27.5	28.0	27.7	27.6	26.2
		MCDB	31.8	33.0	34.2	33.8	33.4	33.4	32.0	31.9	32.4	31.3	31.7	31.1
	2%	WB	25.8	26.5	26.9	27.5	27.5	27.2	26.8	27.0	27.4	27.2	27.0	25.7
		MCDB	31.0	32.2	32.8	33.3	32.9	32.7	31.7	32.0	31.9	31.2	31.4	30.7
	5%	WB	25.2	26.0	26.5	27.1	27.1	26.9	26.4	26.6	27.0	26.9	26.5	25.2
		MCDB	30.1	31.6	32.1	32.9	32.6	32.2	31.3	31.4	31.6	31.0	30.6	30.0
	10%	WB	24.7	25.5	26.1	26.7	26.8	26.5	26.1	26.2	26.6	26.6	26.0	24.6
		MCDB	29.0	30.7	31.4	32.5	32.2	31.4	30.8	30.6	30.9	30.4	29.6	29.1
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.8	10.4	10.0	8.7	7.7	7.2	7.4	7.4	7.0	7.7	9.2
		MCDBR	11.5	11.7	11.9	11.3	10.2	9.1	8.5	8.4	8.6	8.0	8.8	10.3
	5% WB	MCWBR	4.5	3.5	2.8	2.4	2.4	2.3	2.3	2.4	2.6	2.6	3.0	3.8
		MCWBR	10.5	10.5	10.4	10.2	9.0	8.2	7.7	7.9	7.9	7.5	7.9	9.6
Clear-Sky Solar Irradiance	taub	0.536	0.605	0.651	0.574	0.490	0.448	0.440	0.441	0.468	0.575	0.521	0.514	
	taud	1.953	1.816	1.738	1.963	2.259	2.375	2.397	2.382	2.297	1.949	2.073	2.031	
	Ebn at Noon	761	727	707	762	817	843	851	860	842	742	770	768	
	Edh at Noon	181	216	238	190	139	122	120	123	134	186	159	164	
All-Sky Solar Radiation	RadAvg	4.81	5.28	5.53	6.01	5.49	5.08	4.76	4.73	4.81	4.66	4.76	4.69	
	RadStd	0.29	0.22	0.41	0.34	0.44	0.36	0.33	0.34	0.32	0.29	0.31	0.21	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	+0.21	N/A	N/A	N/A	N/A	+0.26	+0.28	N/A	N/A	+100	+100

Nomenclature: See separate page